PROBLEM 1.9

Logistic Population Growth Model (Verhulst)

Original statement · ProblemsCh1.pdf, p. 2

Partial Fractions • Carrying Capacity • Malthusian Limit

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: A refinement over Malthus' population growth model discussed in Chapter 1 is provided by the logistic equation:

$$\frac{dp}{dt} = kp\left(1 - \frac{p}{M}\right)$$

where $p(t)$ is the population size at time t, k is its growth rate (difference between birth rate and death rate), and M is the maximum sustainable population (carrying capacity).

Solve the equation for a generic initial condition $p(t_0) = p_0$ and discuss the evolution of the population for $t \to \infty$, comparing with Malthus' ($M \to \infty$) prediction.

PHASE 2: FUNDAMENTAL FORMULATION

First-order nonlinear autonomous ODE: the logistic equation that the class notes solve as their example of a separable equation [Notes, p. 3]. The solution below follows their steps: constant solutions by Picard's theorem, separation, the partial fractions $\frac{1}{u(1-u/M)} = \frac{1}{u} + \frac{1/M}{1-u/M}$, and isolation of p through $E = \frac{p}{M-p}$.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: Constant solutions and separation. By Picard's theorem a solution that takes the value 0 or M is constant: $p \equiv 0$ or $p \equiv M$, or p never reaches these values [Notes, p. 3]. Take $p_0 > 0, p_0 \neq M$; then $p(t) > 0$ and $M - p(t)$ keeps the sign of $M - p_0$. Separating:

$$\int_{p_0}^{p} \frac{du}{u\left(1-\frac{u}{M}\right)} = \int_{t_0}^{t} k\,d\tau = k(t-t_0) \tag{1.9.1}$$

Step 2: Partial fractions (as in the notes):

$$\frac{1}{u\left(1-\frac{u}{M}\right)} = \frac{A}{u} + \frac{B}{1-\frac{u}{M}} \implies 1 = A\left(1-\frac{u}{M}\right) + Bu$$

At $u=0$: $A=1$. At $u=M$: $1=BM$, so $B=\frac{1}{M}$. Multiplying top and bottom of the second fraction by M:

$$\frac{1}{u\left(1-\frac{u}{M}\right)} = \frac{1}{u} + \frac{1/M}{1-u/M} = \frac{1}{u} + \frac{1}{M-u} \tag{1.9.2}$$

Step 3: Definite integration. An antiderivative of $\frac{1}{u}$ ($u>0$) is $\ln u$; for $\frac{1}{M-u}$ substitute $w=M-u$, $dw=-du$: $\int \frac{du}{M-u} = -\ln|M-u|$. By Barrow's rule

$$\int_{p_0}^{p} \frac{du}{u} + \int_{p_0}^{p} \frac{du}{M-u} = \ln\frac{p}{p_0} + \ln\frac{|M-p_0|}{|M-p|} = \ln\frac{p\,(M-p_0)}{p_0\,(M-p)}$$

(the moduli cancel because $M-p$ and $M-p_0$ have the same sign).

Step 4: Exponentiation and isolation of $p(t)$. Equating to $k(t-t_0)$ and exponentiating, with the abbreviation of the notes:

$$\frac{p}{M-p} = \frac{p_0\,e^{k(t-t_0)}}{M-p_0} =: E \implies p = (M-p)E \implies p\,(1+E) = ME$$

$$p(t) = \frac{ME}{1+E} = \frac{Mp_0e^{k(t-t_0)}}{(M-p_0)+p_0e^{k(t-t_0)}}$$

Dividing numerator and denominator by $e^{k(t-t_0)}$:

$$\mathbf{p(t) = \frac{Mp_0}{p_0 + (M-p_0)e^{-k(t-t_0)}}} \tag{1.9.3}$$

Step 5: Asymptotic Limits:

- As $t \to \infty$: $\lim_{t\to\infty} e^{-k(t-t_0)} = 0 \implies \lim_{t\to\infty} p(t) = \frac{Mp_0}{p_0+0} = \mathbf{M}$.
- As $M \to \infty$ (Recovery of Malthus): Dividing numerator and denominator of (1.9.3) by M:

$$p(t) = \frac{p_0}{\frac{p_0}{M} + \left(1 - \frac{p_0}{M}\right) e^{-k(t-t_0)}} \implies \lim_{M\to\infty} p(t) = \frac{p_0}{0 + (1-0)e^{-k(t-t_0)}} = \mathbf{p_0 e^{k(t-t_0)}}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$p(t) = \frac{Mp_0}{p_0 + (M - p_0)e^{-k(t-t_0)}}$$

$$\lim_{t\to\infty} p(t) = M, \quad \lim_{M\to\infty} p(t) = p_0 e^{k(t-t_0)}$$

Physical Interpretation: The logistic curve exhibits a sigmoidal shape when $p_0 < M/2$, passing through an inflection point of maximum growth rate at $p = M/2$.